Appendix: Matrix Calculus for the Normal Equations

INFO 521 · Module 1 · Lecture B — supplement

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Where we are

Lecture B stated the gradient of the least-squares loss directly:

$$\nabla_{\mathbf{w}}\mathcal{L} = -\frac{2}{N}\,\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\mathbf{w}).$$

This appendix fills in the two vector-calculus identities behind it and turns the crank to $\hat{\mathbf{w}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$.

Two identities

For a constant vector $\mathbf{a}$ and constant matrix $\mathbf{A}$:

$$\frac{\partial\left(\mathbf{a}^{\top}\mathbf{w}\right)}{\partial\mathbf{w}} = \mathbf{a}$$

$$\frac{\partial\left(\mathbf{w}^{\top}\mathbf{A}\mathbf{w}\right)}{\partial\mathbf{w}} = \left(\mathbf{A} + \mathbf{A}^{\top}\right)\mathbf{w} \;=\; 2\mathbf{A}\mathbf{w} \quad (\mathbf{A}\ \text{symmetric}).$$

Expand the loss

Write the loss as a quadratic in $\mathbf{w}$:

$$\mathcal{L}(\mathbf{w}) = \frac{1}{N}(\mathbf{y} - \mathbf{X}\mathbf{w})^\top(\mathbf{y} - \mathbf{X}\mathbf{w})$$

Multiply out — the two cross terms are equal scalars ($\mathbf{y}^\top\mathbf{X}\mathbf{w} = \mathbf{w}^\top\mathbf{X}^\top\mathbf{y}$):

$$\mathcal{L}(\mathbf{w}) = \frac{1}{N}\left(\mathbf{y}^\top\mathbf{y} - 2\,\mathbf{w}^\top\mathbf{X}^\top\mathbf{y} + \mathbf{w}^\top\mathbf{X}^\top\mathbf{X}\mathbf{w}\right)$$

Differentiate & solve

Apply the two identities term by term ($\mathbf{a} = \mathbf{X}^\top \mathbf{y}$, $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ symmetric):

$$\nabla_{\mathbf{w}} \mathcal{L} = \tfrac{1}{N}\left(-2\,\mathbf{X}^\top \mathbf{y} + 2\,\mathbf{X}^\top \mathbf{X}\mathbf{w}\right) = -\tfrac{2}{N}\,\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\mathbf{w})$$

Set it to zero — the $-2/N$ drops out:

$$\mathbf{X}^\top \mathbf{X}\,\hat{\mathbf{w}} = \mathbf{X}^\top \mathbf{y} \qquad \Longrightarrow \qquad \boxed{\hat{\mathbf{w}} = \left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top \mathbf{y}}$$